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Research

A MADS-box gene with similarity to *FLC* is induced by cold and correlated with epigenetic changes to control budbreak in kiwifruit

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Summary

- Temperate perennials require exposure to chilling temperatures to resume growth in the following spring. Growth and dormancy cycles are controlled by complex genetic regulatory networks and are governed by epigenetic mechanisms, but the specific genes and mechanisms remain poorly understood.
- To understand how seasonal changes and chilling regulate dormancy and growth in the woody perennial vine kiwifruit (*Ac*, *Actinidia chinensis*), a transcriptome study of kiwifruit buds in the field and controlled conditions was performed. A MADS-box gene with homology to *Arabidopsis FLOWERING LOCUS C* (*FLC*) was identified and characterized.
- Elevated expression of *AcFLC-like* (*AcFLCL*) was detected during bud dormancy and chilling. A long noncoding (lnc) antisense transcript with an expression pattern opposite to *AcFLCL* and shorter sense noncoding RNAs were identified. Chilling induced an increase in trimethylation of lysine-4 of histone H3 (H3K4me3) in the 5′ end of the gene, indicating multiple layers of epigenetic regulation in response to cold. Overexpression of *AcFLCL* in kiwifruit gave rise to plants with earlier budbreak, whilst gene editing using CRISPR-Cas9 resulted in transgenic lines with substantially delayed budbreak, suggesting a role in activation of growth.
- These results have implications for the future management and breeding of perennials for resilience to changing climate.

Introduction

In temperate woody perennials, winter dormancy is necessary to survive unfavourable winter conditions and is required for synchronized budbreak and flowering in the following spring (Cooke *et al.*, 2012; Brunner *et al.*, 2017). It is a dynamic process of no visible growth, divided into paradormancy, endodormancy and ecodormancy phases, with endodormancy representing the deepest, true dormancy, regulated by endogenous factors that prevent meristem outgrowth at permissive external conditions (Lang *et al.*, 1987).

Dormancy onset is a gradual process including bud set, acclimation to cold and establishment of endodormancy. Release from dormancy requires accumulation of winter chilling (Lang *et al.*, 1987; Rohde *et al.*, 2007; Brunner *et al.*, 2017), enabling growth resumption in spring. Each of these steps are associated with transcriptional and metabolic changes, with multiple regulatory programmes controlling onset, maintenance and release from dormancy (Ruttink *et al.*, 2007; Maurya & Bhalerao, 2017; Beauvieux *et al.*, 2018; Lloret *et al.*, 2018; Liu & Sherif, 2019). Central to these regulatory programmes are MADS-box genes. Tree homologs of *Arabidopsis thaliana* (Arabidopsis) floral meristem identity genes *APETALA* (*AP1*) and *FRUITFULL* (*FUL*) affected growth cessation in poplar (Hoenicka *et al.*, 2008; Azeez *et al.*, 2014) and promoted maturity in citrus, apple and birch (Peña *et al.*, 2001; Elo *et al.*, 2007; Flachowsky *et al.*, 2007). Homologs of Arabidopsis floral repressor *SHORT VEGETATIVE PHASE* (*SVP*), including *DORMANCY ASSOCIATED MADS-box* (*DAM*) genes, were identified as major determinants of chilling requirement in some fruit trees (Fan *et al.*, 2010; Calle *et al.*, 2019). They act as functionally conserved growth repressors central to multiple aspects of dormancy regulation (Sasaki *et al.*, 2011; Wu *et al.*, 2017a,b, 2018; Singh *et al.*, 2018, 2019; Azeez *et al.*, 2021). Homologs of *SUPPRESSOR OF OVEREXPRESSION OF CO1* (*SOC1*) affected the chilling requirement for dormancy break, possibly through interactions with *SVP* (Trainin *et al.*, 2013; Voogd *et al.*, 2015; Wu *et al.*, 2019).

Genes with homology to Arabidopsis *FLOWERING LOCUS C* (*FLC*) have been studied for their roles in phenology and

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regulation of perennial habit (Wang, 2009). FLC is a major floral repressor acting in the vernalization pathway to inhibit expression of *FLOWERING LOCUS T* (*FT*) via direct binding to the CArG motifs in the *FT* locus (Simpson & Dean, 2002; Searle *et al.*, 2006). Expression of *FLC* is regulated at the epigenetic level and involves noncoding RNA (ncRNA) and chromatin modifications; vernalization promotes expression of long noncoding (lnc) antisense transcripts across the *FLC* locus, collectively termed *COOLAIR* (Ietswaart *et al.*, 2012), a sense lncRNA expressed from the long first intron of *FLC*, named *COLDAIR* (Heo & Sung, 2011; Csorba *et al.*, 2014), and a lncRNA termed *COLDWRAP*, transcribed from the *FLC* promoter (Kim & Sung, 2017; Kim *et al.*, 2017). These transcripts reduce accumulation of the *FLC* transcript by recruiting protein complexes that impact on *FLC* chromatin states. Vernalization induces modifications of the histones present at the *FLC* locus, most notably trimethylation of lysine-27 of histone H3 (H3K27me3) (Shindo *et al.*, 2006), commonly associated with repressed chromatin states. Conversely, high expression of *FLC* is associated with elevated trimethylation of lysine-4 of histone H3 (H3K4me3), a histone mark associated with high transcription (Sims *et al.*, 2003).

The presence of functional *FLC* orthologs outside the Brassicaceae has not been widely demonstrated. However, genes with sequence homology to *FLC* and its paralogs *MADS AFFECTING FLOWERING* (*MAF*) genes have been identified and studied. At least two genes with homology to *FLC* have been reported in apple (Kumar *et al.*, 2016; Kagaya *et al.*, 2020) and grape (Díaz-Riquelme *et al.*, 2009), one in peach (Wells *et al.*, 2015), one in pear (Niu *et al.*, 2015), and at least four in poplar (Leseberg *et al.*, 2006; Chen *et al.*, 2018). In apple, an *FLC* with elevated expression at dormancy release was associated with a major quantitative trait locus for the timing of budbreak (Porto *et al.*, 2015). Peak accumulation of apple *MdFLC* coincided with fulfilment of the chilling requirement in high- and low-chill cultivars (Nishiyama *et al.*, 2021). In citrus, seasonal fluctuation in expression of *FLC* homologs was reported (Zhang *et al.*, 2009; Y.-J. Ma *et al.*, 2020), suggestive of negative regulation of citrus flowering. Furthermore, a mandarin *FLC CcMADS19* (Hou *et al.*, 2014; Agustí *et al.*, 2020) acted to repress the citrus *FT* and drive the biennial bearing pattern. *CcMADS19* was epigenetically regulated through different DNA and histone methylation marks which correlated with the presence or absence of fruit (Agustí *et al.*, 2020).

Kiwifruit (*Actinidia* spp.) are temperate woody perennial vines requiring winter dormancy and a period of winter chilling to resume growth in spring (Brundell, 1976). Short days in autumn initiate growth cessation and transition to dormancy (Lionakis & Schwabe, 1984; Snelgar *et al.*, 2007). There is a considerable difference in the chilling requirement among *Actinidia*. Most *Actinidia chinensis* var. *chinensis* achieve maximum budbreak with less than 700 h at temperatures below 7°C, while *Actinidia chinensis* var. *deliciosa* require substantially more chilling (Wall *et al.*, 2008; Kataoka *et al.*, 2014; Wang *et al.*, 2017). Insufficient chilling leads to a sporadic budbreak, low flower numbers and subsequent low fruit yields (Linsley-Noakes & Allan, 1987; Austin *et al.*, 2002). Recent research revealed that MADS-box genes affect development, including the time of budbreak and shoot outgrowth (Voogd *et al.*, 2015; Wu *et al.*, 2017b, 2018) and duration of flower development (Wu *et al.*, 2014). However, the genes and signalling pathways underlying responses to environmental cues, particularly cold, are still largely unknown.

In this study, we performed an RNA-sequencing (RNA-Seq) analysis using kiwifruit axillary shoot buds in the field and under controlled conditions, to monitor transcriptome changes during dormancy onset and chilling-mediated release from dormancy. We identified a gene with homology to *FLC* which responds strongly to cold. We studied the regulation of its expression and functionally characterized the gene using overexpression and mutagenesis using CRISPR-Cas9. This study determines the role of an *FLC*-like gene in kiwifruit phenology, demonstrating functional divergence despite the conserved aspects of *FLC* regulation.

Materials and Methods

Plant material

For the RNA-Seq experiment, kiwifruit plant material was collected from reproductively mature *Actinidia chinensis* var. *chinensis* 'Zesy002'. 'Zesy002' was grafted on *Actinidia chinensis* var. *deliciosa* 'Hayward' rootstocks in August 2010 and the plants were grown in the Plant & Food Research orchard near Te Puke, New Zealand, under natural conditions and according to standard orchard practices. The temperature and chilling accumulation data (Luedeling, 2012) were downloaded from MetWatch Online (http://www.metwatch.co.nz). Shoot axillary bud tissue was collected at monthly intervals from November 2014 to August 2015. All sampling was performed at midday and *c.* 10 buds were collected from three individual plants (biological replicates) at each time point. A subset of canes was used to monitor budbreak. Terminal buds were collected from the same plants in November, January and March. Because of the limited availability of terminal buds, five buds from one or two plants were used per sample. To test gene expression upon exposure to cold, 'Zesy002' canes were collected in autumn, April 2017, defoliated and exposed to 4°C for 36 h, 1, 2, 3 or 4 wk, prior to bud dissection. A minimum of five canes from a total of 10 plants were used per treatment and each cane contained a minimum of eight buds. The canes were stored in plastic bags to prevent moisture loss. A subset of canes was used to monitor budbreak. The low-chill *Actinidia chinensis* var. *chinensis* 'Hort16A' and high-chill *Actinidia chinensis* var. *deliciosa* 'Hayward' bud samples have been described (Varkonyi-Gasic *et al.*, 2013) and the tissue expression was assessed according to Brian *et al.* (2021). For the chromatin immunoprecipitation coupled with quantitative polymerase chain reaction (ChIP-qPCR) experiment, 'Hort16A' samples were described previously (Wu *et al.*, 2019). In addition, 'Zesy002' canes were collected on 3 April 2019 at the Plant & Food Research orchard. The canes were exposed to 4°C and axillary buds were harvested on day 0 before cold treatment, after 1.5 and 30 d of cold treatment. All sampling was performed at midday using buds from three individual plants (biological replicates).

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Forced budbreak assay on excised canes

Dormancy status was determined as described previously (Varkonyi-Gasic *et al.*, 2013), using stem cuttings maintained at room temperature and long day conditions (16 h : 8 h, light : dark) to force budbreak. The canes were collected before the temperature daily minimum dropped below 7°C. A minimum of 10 cuttings per treatment were used and the percentage of cuttings breaking bud and the number of days required for visible budbreak were recorded.

RNA extraction and expression studies

Total RNA was isolated using the Spectrum Plant Total RNA Kit (Sigma-Aldrich, St Louis, MO, USA). Sequencing libraries were constructed using the TruSeq stranded messenger RNA (mRNA) sample preparation method (Illumina, San Diego, CA, USA) and subsequently sequenced by a HiSeq 2500 Sequencing System (Illumina) at Australia Genomics Resource Facility (Brisbane, Australia), obtaining paired-end reads of 125 bp and an average of at least 13 million reads per library. Mapping to the manually curated *Actinidia chinensis* Red5 genome (Pilkington *et al.*, 2018) was performed using STAR v.2.5.3a (Dobin *et al.*, 2012). Uniquely mapped read pairs were counted using HTSEQ v.0.9.1 (Anders *et al.*, 2015) and gene expression was quantified as fragments per kilobase of transcript per million mapped reads (FPKM). The resulting count tables from HTSeq were used for differential gene expression analysis using the R package DESEQ2 v.1.12.2 (Love *et al.*, 2014). Differentially expressed gene (DEG) sets were visualized using Intervene and CLUSTVIS software (Metsalu & Vilo, 2015; Khan & Mathelier, 2017). Gene annotations were obtained with BLAST v.2.6.0 (Altschul *et al.*, 1997) using *Actinidia chinensis* Red5 amino acid sequences querying the Arabidopsis TAIR10_pep database. The gene ontology (GO) term enrichment was performed using the Arabidopsis reference list in the Gene Ontology Resource (http://geneontology.org/).

For reverse transcription (RT)-qPCR expression studies, a 500-ng aliquot of total RNA was treated with DNase I, then reverse-transcribed using Superscript III (Invitrogen, Carlsbad, CA, USA) with oligo-d(T), a mix of oligo-d(T) and random primers (QuantiTect, Qiagen, Hilden, Germany), or a mix of a gene-specific primer with an extension tag and a primer for *ACTIN*, followed by qPCR with primers designed to amplify products between 81 bp and 146 bp in size. The primer sequences are listed in Supporting Information Table S1. The qPCRs were performed using FastStart DNA MasterPLUS SYBR Green I reaction mix on a LightCycler® 1.5 instrument (Roche Diagnostics, Basel, Switzerland). A nontemplate control was included in each run. Amplification was carried out with an initial denaturing step at 95°C for 5 min, then 40–50 cycles of 95°C for 5 s, 60°C for 5 s, and 72°C for 10 s. The PCR efficiency for each individual sample was calculated using the LINREGPCR v.11.1 software (Ramakers *et al.*, 2003; Ruijter *et al.*, 2009). The mean efficiency per amplicon was included in the calculation of relative expression ratios according to the comparative cycle threshold (Ct) method (Pfaffl, 2001). The Ct values were determined using the second derivative maximum method in the LightCycler® 480 software v.1.5.0.

AcFLCL transcript assembly and visualization

Visualization of the mapped read coverage in the *AcFLCL* locus was achieved using the genome browsing application Integrative Genomics Viewer IGV v.2.8.13 (Robinson *et al.*, 2011). Sense and antisense transcripts were identified for each of the RNA-Seq libraries using CUFFLINKS v.2.2.1 (Trapnell *et al.*, 2012), guided by the reference genome (Pilkington *et al.*, 2018). Cufflinks-generated output files in GTF format were then loaded into GENEIOUS v.10.0.9 (Biomatters, Auckland, New Zealand) (https://www.geneious.com) and manually curated.

Phylogeny

Sequence alignment of MADS-box proteins was performed using GENEIOUS MUSCLE alignment (eight iterations) and the phylogenetic tree was built using the Neighbor-Joining method with 1000 bootstrap replicates (GENEIOUS).

Promoter identification and analysis

Up to 3000 bp of sequences upstream of the transcription start site (TSS) were extracted from all 33 318 genes in the reference genome (Pilkington *et al.*, 2018) using SAMTOOLS v.1.9 (Danecek *et al.*, 2021) and scanned for the presence of the CArG box motifs $CC(A/T)_6GG$ using SEQKIT v.0.7.0 (Shen *et al.*, 2016). Statistical significance for over- or under-enrichment for CArG boxes in promoters of subsets of DEGs identified by RNA-Seq was based on the cumulative distribution function of the hypergeometric distribution.

Histone modification experiment

ChIP assays and qPCR detection were performed with three biological replicates, using the previously described method and consumables (Wu *et al.*, 2019) and oligonucleotide primers listed in Table S1.

AcFLCL overexpression and gene editing constructs

For *AcFLCL* overexpression (*35S:AcFLCL*), a clone containing full-length complementary DNA (cDNA) was obtained from the Plant & Food Research *Actinidia* EST library (Crowhurst *et al.*, 2008) and cloned as a *Spe*I-*Xho*I fragment into the pSAK778 binary vector (Drummond *et al.*, 2009). A binary CRISPR gene editing construct was designed based on the multiplex tRNA-gRNA system (Xie *et al.*, 2015) with a modified sgRNA structure (Dang *et al.*, 2015). Suitable targets within the *AcFLCL* locus were identified using the GENEIOUS v.10.0.9 CRISPR selection tool with quality scoring based on Doench *et al.* (2014). Four guides, targeting five locations, were chosen (Table S2). A synthetic gRNA/pre-tRNA fused fragment placed downstream of the Arabidopsis U6-26 promoter between attL Gateway sites

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(pENTR-pAtU6.26, sequence in Table S2) was obtained from General Biosystems (https://www.generalbiosystems.com) and used as a template for amplification of Golden Gate parts (primers in Table S2) as well as the cloning destination of the Golden Gate reaction by replacing the gRNA/pre-tRNA fused fragment in pENTR-pAtU6.26 with the assembled Golden Gate parts (pENTR-pAtU6.26-FLC-g1.g2.g3.g4). PCR was performed using iProof High-Fidelity DNA Polymerase (Bio-Rad, Hercules, CA, USA) with the following reaction conditions: 98°C-30 s, five cycles (98°C-10 s/45°C-30 s/72°C-10 s), 20 cycles (98°C-10 s/55°C-30 s/72°C-10 s), 72°C-5 min. Purified PCR products were digested with *Bsa*I then ligated (T7 DNA Ligase; New England Biolabs, Ipswich, MA, USA) into *Bsa*I-digested pENTR-pAtU6.26. After sequence verification this entry clone was recombined by LR reaction using Gateway cloning into pKRS (Varkonyi-Gasic *et al.*, 2019), resulting in pKRS-pAtU6.26-FLC-g1.g2.g3.g4. This binary vector was subsequently transformed into *Agrobacterium tumefaciens* strain EHA105 by electroporation.

Plant transformation and growing conditions

Agrobacterium-mediated transformation of *Actinidia chinensis* 'Hort16A' was performed as previously described (Wang *et al.*, 2007). Rooted transgenic plants were established in the glasshouse, allowed to overwinter in the growth room with supplemented heating and light (temperature 22°C, 16 h : 8 h, light : dark), then transferred to the glasshouse and grown at ambient conditions. Shoot axillary buds were monitored for budbreak, recorded as the date of leaf tip emergence.

Genotyping of transgenic plants

Genomic DNA was isolated using the DNeasy Plant Mini Kit (Qiagen) and PCR using iProof High-Fidelity DNA Polymerase (Bio-Rad) was used for genomic testing of transgenic plants. For each plant, two PCR fragments covering either exon 3, 4, and 5 (guides 1, 2, and 3) or exons 6, 7 and 8 (guide 4) were amplified (oligonucleotide primers in Table S1). PCR conditions were as follows: 98°C-3 min, 30 cycles (98°C-10 s/64°C-30 s/72°C-50 s), 72°C-5 min. Purified PCR products were subjected to sequence analysis using primers listed in Table S1. Tracking of indels by decomposition (TIDE) software (Brinkman *et al.*, 2014) in combination with manual decomposition were used to determine CRISPR-induced mutations. Natural single nucleotide polymorphisms (SNPs) in the *AcFLCL* locus of the 'Hort16A' genome were used to distinguish between the two alleles.

Results

Cold establishes, then breaks endodormancy in kiwifruit

Kiwifruit *Actinidia chinensis* plants break buds in late winter, followed by rapid shoot elongation and flower development in lower leaf axils. By late spring, buds in the upper leaf axils become clearly visible and enlarge further as shoots continue active growth during the summer months (Fig. 1a). Shortening of the day-length in autumn initiates bud set, followed by leaf-drop and establishment of winter dormancy (Fig. 1a), when temperatures below 7°C start contributing to winter chilling accumulation (Figs 1b, S1). Exposure to low temperature is required both to fully establish and subsequently break endodormancy. Canes sampled in autumn and subjected to cold treatment (4°C) for 36 h were not capable of breaking buds, but increasing

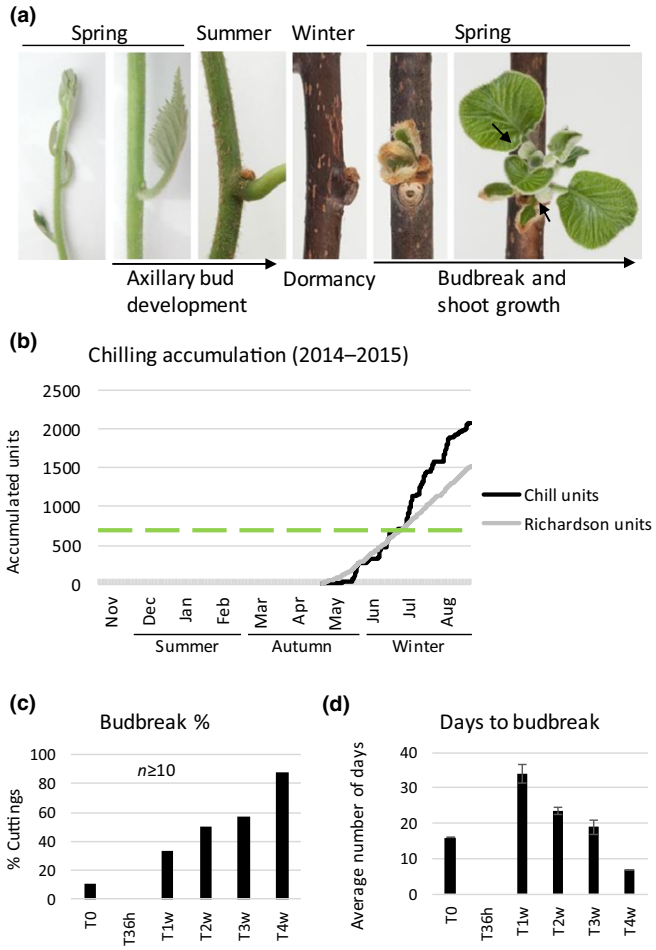


Fig. 1 Sampling of axillary buds representing different stages of growth and dormancy cycles. (a) The lifecycle of an *Actinidia chinensis* axillary shoot bud: the latent bud in the upper leaf axil enlarges over spring and summer months, becomes dormant in the winter and gives rise to a spring shoot with flower buds developing in lower leaf axils (arrows). (b) Accumulation of winter chilling during the season at the location of bud sampling. Accumulation of chilling was presented as chill units, assigning increasing amount of chilling hours for every full hour at temperatures below 7°C and excluding hours below 0°C, or Richardson units, assigning different weight to different temperature bands and recognizing the negative chill contribution of high temperatures. (c, d) Single node cutting assay was performed with canes sampled before the temperature daily minimum dropped below 7°C. A single node cutting sample was taken immediately (T0) and the remaining canes were maintained at 4°C and single node cuttings were taken after 36 h (T36h), 1 wk (T1w), 2 wk (T2w), 3 wk (T3w) and 4 wk (T4w), immersed in water and monitored for budbreak. A minimum of 10 cuttings per treatment were used and the percentage of cuttings breaking bud was recorded (c) with the number of days required for visible budbreak in those cuttings being represented as a mean ± SE (d).

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numbers of buds broke with longer chilling periods (Fig. 1c). The average time to budbreak and the duration of chilling were negatively correlated (Fig. 1d), demonstrating a decrease of the depth of dormancy and transition to ecodormancy with accumulation of chilling.

Transcriptional regulation of bud growth, dormancy and cold responses

To study the transcriptional regulation of growth and dormancy cycles, buds collected from field-grown plants at monthly intervals from November to August were subjected to RNA-Seq analysis. A clear separation between sampling points representing axillary bud phenological stages and actively growing terminal buds was identified (Fig. S2a). A large number of DEGs were associated with developmental transitions occurring between sampling times (Fig. S2b; Table S3). Growth cessation and dormancy onset occurring in autumn, between March and May, were characterized by large increases in transcription of genes associated with response to oxidative stress, temperature stress and water deprivation, such as *CATALASE 2*, genes encoding for heat shock proteins, and the late embryogenesis abundant (LEA) protein family. At the same time, a large decrease in transcription of cell-wall related genes and developmental regulators, including *CENTRORADIALIS* (*CEN*), was observed (Table S4). Bud transcriptomes reflecting winter chilling (May–July) showed an increase in photosynthesis-related genes and developmental regulators such as *AcFT*, *PHYTOCHROME INTERACTING FACTOR* (*PIF*) and two MADS-box gene *FUL* homologs (Table S4).

To further study the response to chilling, axillary buds collected from canes exposed to 4°C for up to 4 wk were used. The sampling times corresponded to cold-mediated establishment of endodormancy (T36h) and accumulation of winter chilling (T1w–T4w). This was supported by results of the budbreak assay (Fig. 1c,d), as well as by the clear separation between T0, T36h and T1w–T4w sampling points (Fig. S2c), accompanied by a gradual reduction in numbers of DEGs (Fig. S2d; Table S3). The short cold treatment (T0–T36h) gave rise to expression of genes associated with response to light, cold and osmotic stress, accumulation of osmoprotectants and upregulation of pathways sequestering iron and nitrogen. Even at this early stage, accumulation of transcripts implicated in photomorphogenesis and flowering-related B-box proteins, multiple *PSEUDO RESPONSE REGULATOR 5* (*PPR5*) homologs and cycling DOF factors was observed. Further exposure to cold (T36h–T4w) resulted in increased accumulation of stress response genes and changes in cell wall modification factors (Table S5).

Many commonalities in upregulated and downregulated genes identified in bud transcriptomes during autumn (March–May) and following a short cold treatment (T0–T36h) (Fig. 2a; Table S6) demonstrated the role of multiple stress and defence pathways in dormancy onset, accompanied by cessation of development, patterning and reduced photosynthesis (Fig. S3a,b). Similarly, a comparison of DEGs during winter (May–July) and in response to prolonged cold exposure (T36h–T4w) revealed common features in the regulation of winter chilling (Fig. 2a; Table S6). Expression of the upregulated transcripts was mostly specific to the winter season and exposure to cold, with little expression detected in axillary buds during active growth (November–April), in T0 samples or in terminal buds (Fig. 2b). In contrast, many of the DEGs identified as downregulated over the dormancy and chilling period were expressed in actively growing terminal buds (Fig. 2b; Table S6), supporting their role in active growth. In addition to changes in cell wall-related categories, dormancy release was accompanied by a reduction in stress and defence response transcripts (Fig. S3c).

In addition to the common DEGs (Table S6), many DEGs identified in bud transcriptomes during winter (May–July, Table S4) were absent from buds exposed to cold (T36h–T4w, Table S5). The upregulated genes in this set demonstrated early signs of growth and photosynthesis resumption specific for field-grown plants (Fig. S4a), not yet occurring in buds exposed to constant cold. Finally, DEGs specific for bud samples in August reflected the transition to active growth occurring after accumulation of sufficient chilling. They were highly enriched in cell division and expansion genes coinciding with budbreak (Table S7; Fig. S4b).

An *FLC*-like gene is upregulated in buds during dormancy and winter chilling

Eleven genes were consistently upregulated from dormancy onset to release in the field as well as controlled chilling conditions (Fig. 2a), suggesting important roles in regulation of bud development and physiology. They included regulators of development, hormone metabolism, RNA processing and protein ubiquitination (Fig. 3a). A MADS-box gene (Acc05562) was one of the 11 genes, showing an initial increase in expression during dormancy onset, followed by a very large increase over winter months, as well as gradual upregulation in canes experiencing artificial chilling. Increased expression in buds during winter and with artificial chilling was also detected in other kiwifruit cultivars regardless of their winter-chilling requirement (Fig. 3b,c), indicating a conserved role in bud responses to cold. This increased expression preceded accumulation of transcripts such as *FUL* and *CYCLIN B*, marking growth, and *PISTILLATA*, an indicator of flower development (Fig. 3b). The function in bud responses was corroborated by the enrichment of MADS-box protein-binding CArG motifs in proximal promoter regions of genes upregulated during winter (Table S8), including *FUL* homologs, a cytokinin metabolism gene *CYP735A*, and B-box transcription factors (Tables S4–S6). Underrepresentation of CArG boxes coincided with decreased expression of Acc05562 at budbreak (Table S8).

Phylogenetic analysis of predicted kiwifruit MADS-box proteins annotated in the *Actinidia chinensis* genome (Fig. S5) placed Acc05562 and a homolog Acc14299 in the same clade with Arabidopsis FLC and MAF proteins. Further manual interrogation of the *Actinidia chinensis* genomic sequence identified another homolog, almost identical to a gene previously annotated *Actinidia deliciosa AdFLC* (Ruelens *et al.*, 2013), to which we assigned

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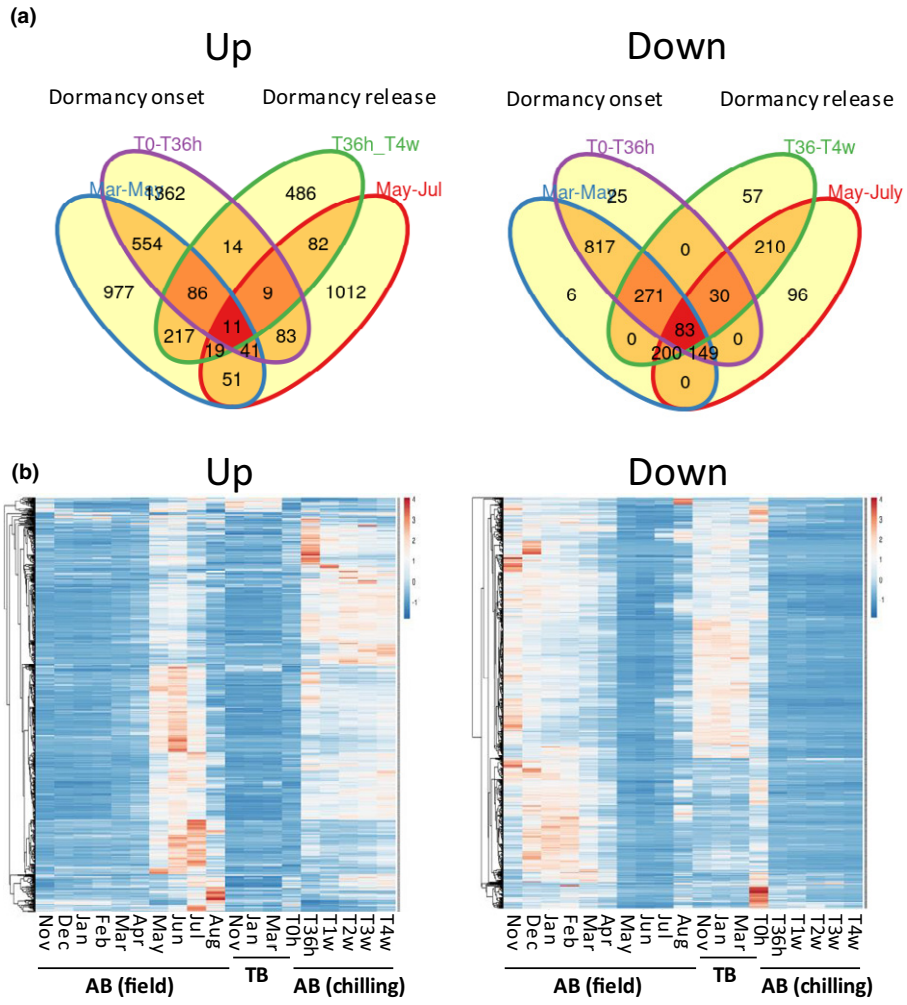


Fig. 2 Changes in bud transcriptomes ($P_{adj} < 0.05$, fold change ≥ 2) at dormancy onset and release for *Actinidia chinensis*. (a) Specific and common sets of genes upregulated (UP) and downregulated (DOWN) during autumn and with a short cold treatment (dormancy onset), or in winter and with longer chilling (dormancy release). A small set of 11 differentially expressed genes (DEGs) was consistently upregulated. (b) Expression and correlation of upregulated (UP) and downregulated (DOWN) genes. AB, axillary buds; TB, terminal buds.

the gene identification number Acc33776. The Acc05562-predicted protein of 242 amino acids is encoded by a gene with eight exons, in which exons 6 and 7 are identical. Acc14299 and Acc33776 have a similar gene structure, but with only seven exons and different intron lengths (Fig. 3d). The predicted coding regions of all three genes show high similarity in the N-terminal region corresponding to the MADS-box, but less similarity in the rest of the sequence compared with Arabidopsis FLC and MAF proteins (Fig. S6). Blast query suggested similarity to other MADS-box proteins, particularly FUL. Unlike the characterized kiwifruit homologs of FUL (Varkonyi-Gasic *et al.*, 2011), none of the FLC-like predicted proteins contained the FUL signature C-terminal sequence L/MPPWML, and all three were members of a separate subclade related to Arabidopsis FLC/MAF (Figs 3e, S5). Therefore, for further study we refer to Acc05562 as *AcFLC-like* (*AcFLCL*).

Differential expression of coding and noncoding RNAs in the *AcFLCL* locus

Examination of reads mapped to the *AcFLCL* locus (Fig. 4a) identified sense transcripts corresponding to *AcFLCL* exons, with additional shorter sense RNA transcripts mapping to the first exon and the beginning of the first intron. Reads across all exons were detected during dormancy and chilling, but the shorter transcripts were also detected before dormancy. Additionally, antisense transcripts mostly confined to the distal end of the *AcFLCL* locus were identified. Peak accumulation of sense transcripts coincided with dormancy and accumulation of chilling. In contrast, the antisense transcript accumulation declined during winter months and in buds exposed to cold (Fig. 4a).

Manual curation of RNA-Seq reads identified different variants of *AcFLCL* transcripts and multiple forms of the shorter sense transcript and the antisense transcripts. The presence and differential accumulation of these various RNAs was confirmed using RT-PCR with specific oligonucleotide primers (Fig. 4b). While the overall accumulation of antisense transcripts declined towards the end of winter, at this time distal antisense transcripts could be detected (Fig. 4b). Neither of the antisense or shorter sense variants was capable of encoding a long peptide, suggesting they acted as ncRNAs.

Quantification of sense and antisense reads was further used to confirm the contribution of *AcFLCL* to chilling responses in buds (Fig. 4c). Expression of sense *AcFLCL* was clearly associated with chilling, with maximum transcript accumulation in the coldest month of the year (July) and a steady increase over the 4-wk

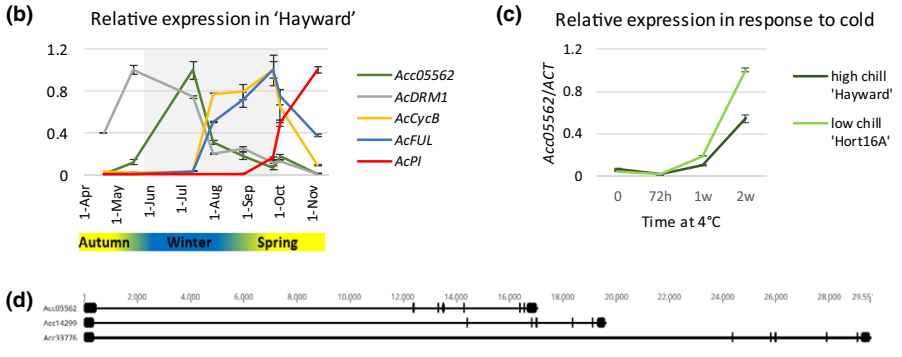
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(a)

gene ID	Homolog	Description/ Predicted function	'Zesy002' axillary bud average FPKMs														
			Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	T0h	T36h	T1w	T2w	T3w	T4w
Acc05562	AT5G65070	MADS-box transcription factor/ Flowering	12.33	18.33	21.33	23.18	28.24	38.96	157.08	417.88	519.78	105.08	28.53	68.82	63.61	99.63	147.72	189.77
Acc26605	AT1G25440	BBX transcription factor/ Development	0.00	0.05	0.08	0.17	0.18	0.15	0.74	3.55	1.96	0.57	0.03	0.26	0.69	1.26	1.39	3.82
Acc09717	AT1G05010	ACO4/ Ethylene biosynthesis	33.45	14.63	11.35	11.78	23.12	28.01	164.60	321.24	370.11	95.73	64.47	133.71	526.34	1028.3	879.88	869.5
Acc13989	AT4G17720	BPL1/ rRNA processing	2.16	1.39	1.43	1.69	2.27	3.02	5.74	19.00	17.76	9.13	1.18	2.38	6.24	20	41.51	49.43
Acc28117	AT4G17090	BAM3/ Chloroplast beta-amylase	0.45	0.32	0.13	0.31	0.37	1.01	40.58	229.32	119.16	8.16	1.3	53.49	56.61	170.75	164.58	260.4
Acc28118	AT4G17090	BAM3/ Chloroplast beta-amylase	0.23	0.19	0.05	0.15	0.29	0.66	17.73	93.49	70.35	4.58	0.52	25.21	34.76	55.63	68.08	113.76
Acc19919	AT1G62770	Plant invertase/ pectin methylesterase inhibitor	1.65	2.55	3.46	6.00	11.04	44.38	149.00	237.20	374.59	308.60	22.07	45.15	46.8	78.81	95.75	122.36
Acc00684	AT5G11090	Serine-rich protein-related	0.00	0.09	0.65	0.38	0.35	1.87	32.42	73.61	69.34	3.66	1.03	10.06	5.51	23.91	22.13	37.94
Acc25785	AT2G30220	GDSL-like Lipase/ Acylhydrolase family protein	0.00	0.00	0.03	0.04	0.02	0.44	3.78	12.85	10.18	0.04	0.02	0.9	4.04	9	17.72	19.1
Acc00747	AT5G52450	MATE efflux family protein/ Stress response	8.32	3.92	4.13	4.05	3.61	6.15	17.59	42.92	62.46	36.61	7.1	20.07	28.23	36.49	56.07	54.14
Acc19679	AT1G02610	RING/FYVE/PHD zinc finger/ Protein ubiquitination	1.37	0.31	0.14	0.83	1.29	0.72	6.95	14.37	20.58	2.87	0.6	2.51	2.44	3.64	4.51	6.91



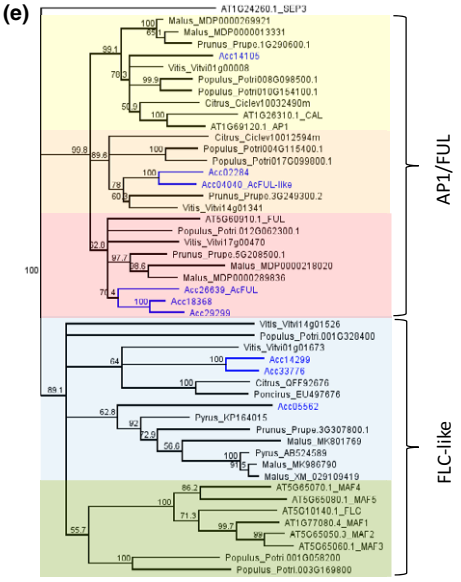


Fig. 3 Identification of kiwifruit *FLC*-like genes. (a) Eleven genes consistently upregulated with dormancy and chilling. (b) Relative expression in high-chill *Actinidia chinensis* var. *deliciosa* 'Hayward' axillary buds collected at monthly intervals during the growth and dormancy cycle. *Actinidia DORMANCY-ASSOCIATED PROTEIN 1* (*AcDRM1*) and the cell-cycle marker *CYCLIN B* (*AcCycB)* were used as markers for dormancy and cell division, respectively. Kiwifruit *FRUITFUL* (*AcFUL*) and *PISTALLATA* (*AcPI*) mark growth and floral organ development, respectively. The shading indicates the period with no visible growth. (c) Relative expression in buds exposed to cold. Expression (b, c) was normalized against *ACTIN* and expressed relative to the sample with highest expression (arbitrary assigned the value one). Error bars represent ± SE of three biological replicates. (d) Genomic structure of kiwifruit *FLC*-like genes. Exons are presented as black boxes. (e) Phylogenetic tree of FLOWERING LOCUS C (FLC) and APETALA1/FRUITFUL (AP1/FUL) predicted proteins. Sequence alignment was performed using Geneious MUSCLE alignment and the phylogenetic tree was built using the Neighbor-Joining method with 1000 bootstrap replicates and *Arabidopsis* SEPALLATA3 (SEP3) as outgroup. Subclusters are indicated with different colours and kiwifruit sequences are given in blue font.

chilling period. The peak accumulation rates were much higher than those detected in other kiwifruit tissues. Inversely correlated antisense expression was observed before chilling and during the spring and summer, with a decline in winter and an increase in August. The peak antisense expression (FPKM > 12) was comparably lower than the sense expression. The homologs Acc14299 and Acc33776 had generally lower expression levels, with a small peak in July and little change in sense transcript accumulation during chilling (Fig. 4d,e). Little or no antisense transcript accumulation (FPKM < 1) and no inverse correlation with sense expression patterns were detected. Similarly, negligible antisense transcript and no correlation with sense expression were observed with other MADS-box genes that showed differential expression in buds over the season (Fig. S7), demonstrating that the potential antisense-mediated regulation is highly specific to *AcFLCL*, rather than a general feature of kiwifruit MADS-box genes expressed in dormant buds.

Histone modifications correlate with *AcFLCL* expression

The sequence homology and presence of ncRNAs indicated that *AcFLCL* may display conservation of other aspects of epigenetic changes described for Arabidopsis *FLC*, such as histone modifications. To test if changes in tri-methylation of lysine residues of histone 3 (H3) may be able to contribute to regulation of *AcFLCL* expression, primers designed in the proximity of the translational start and stop sites and across the first intron were used in ChIP-qPCR experiments (Fig. 5a). An increase in H3K4me3 was detected in the region around the translation start site in the dormant mid-winter buds, correlating with increased *AcFLCL* sense expression. In the distal region of the locus, slightly elevated H3K4me3 was observed before budbreak (Fig. 5b,c).

Comparing the samples before chilling and after chilling for 36 h and 30 d, a dynamic change in H3K4me3 was observed. Short cold exposure resulted in accumulation of H3K4me3, with a peak at the primer 4 position. Longer chilling induced increased accumulation, spread across the upstream region of the locus, corresponding to elevated *AcFLCL* sense transcript (Fig. 5d,e). In this set of samples, little difference in H3K4me3 accumulation was detected in the rest of the locus and no differential accumulation of H3K27me3 could be detected (Fig. 5d,f).

AcFLCL promotes budbreak

For functional characterization, the *AcFLCL* coding region driven by the *35S* promoter was introduced into *Actinidia chinensis* 'Hort16A' and eight lines with high *AcFLCL* expression were generated (Fig. 6a). Forced budbreak assays using canes collected in early winter (30 June) suggested earlier budbreak in *AcFLCL* overexpression lines (Fig. 6b,c). Earlier budbreak was also seen in the glasshouse-grown plants, which produced shoots about a month before the visible budbreak in control plants (Fig. 6d).

To further evaluate the role of *AcFLCL*, gene editing targeting sequences in exons 3, 4, and 5 (g1, g2 and g3) and in the

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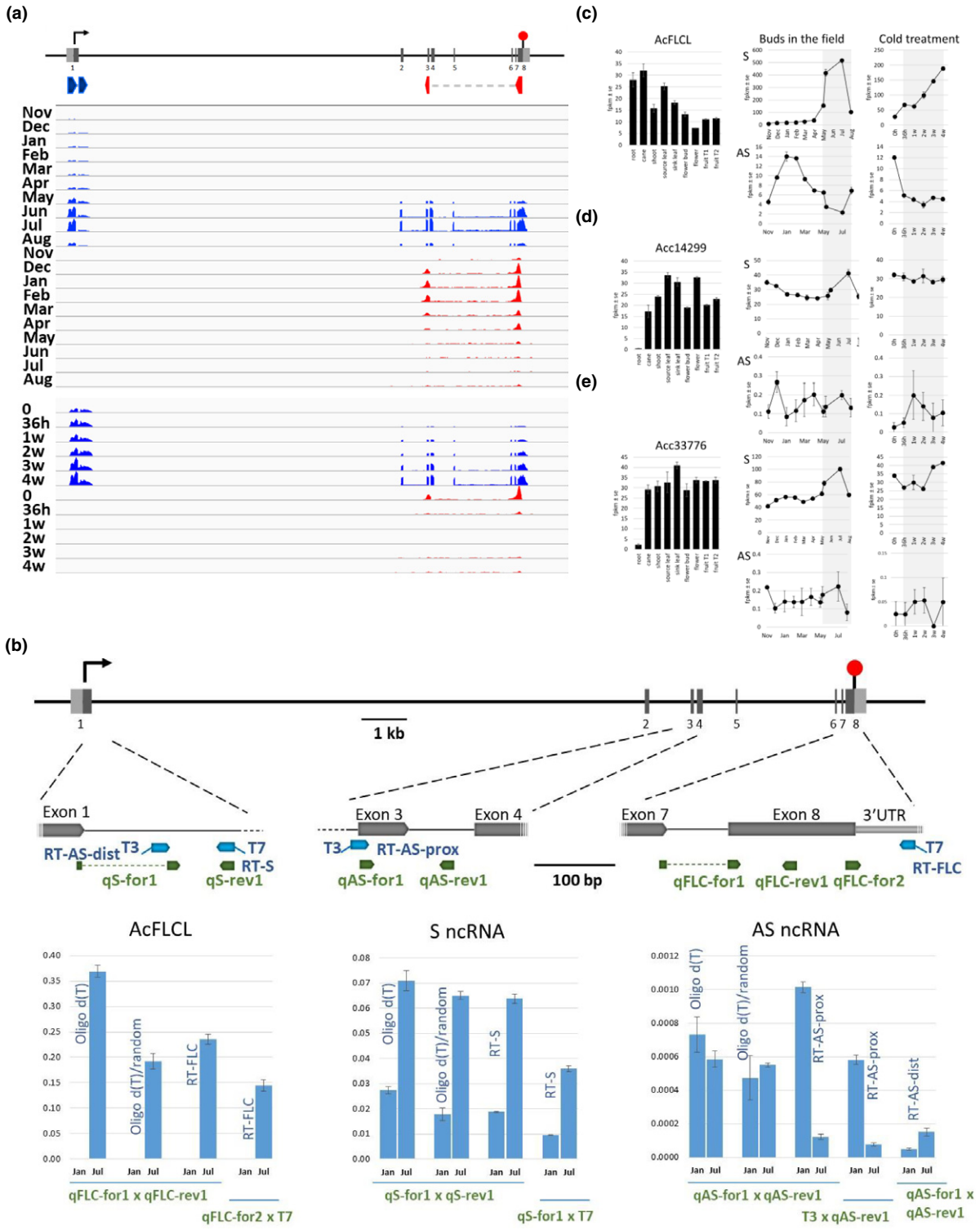


Fig. 4 *AcFLCL* is induced by cold and associated with noncoding RNAs (ncRNAs). (a) Mapping of sense and antisense reads to the *AcFLCL* locus in kiwifruit buds collected from November to August and before and during a cold-treatment. The scale was 4700 for sense (blue) and 350 for antisense (red) reads for the field experiment and 2100 for sense (blue) and 400 for antisense (red) reads for the chilling experiment. The gene model with exons (boxes), translation start and stop sites and the sense (blue) and antisense (red) ncRNAs is presented earlier. (b) Detection of RNAs in the *AcFLCL* locus by reverse transcription quantitative polymerase chain reaction (RT-qPCR). The expression was normalized to kiwifruit *ACTIN* and presented as a mean ± SE of three biological replicates. The location of oligonucleotide primers is given below the gene model. The first-strand RT oligonucleotide and amplification primers are indicated beside and below the graphs, respectively. S, sense; AS, antisense. (c–e) Expression of kiwifruit *FLC*-like genes in axillary buds and different tissues, presented as average fragments per kilobase of transcript per million mapped reads (FPKM) ± SE of three biological replicates. S, sense; AS, antisense.

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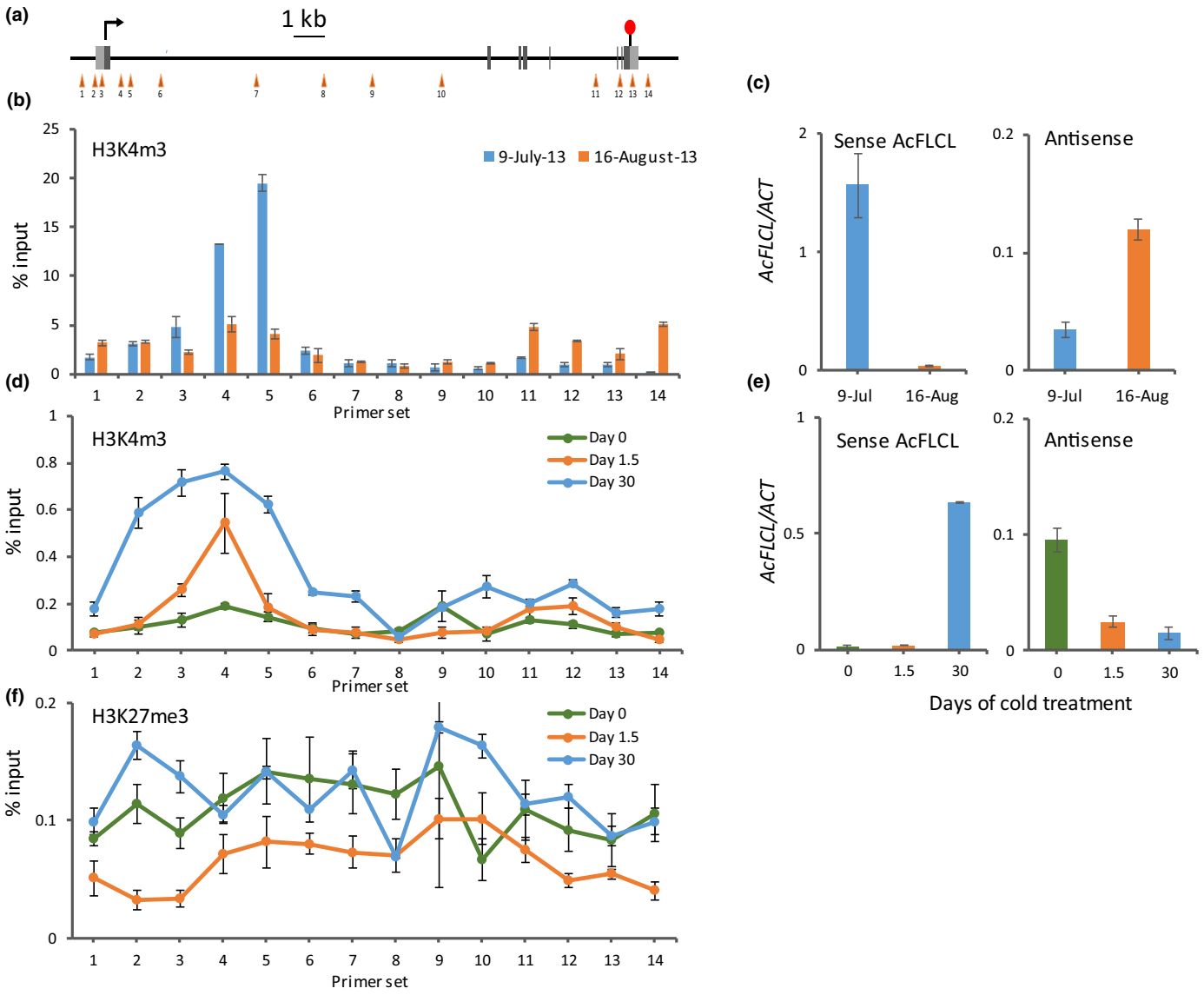


Fig. 5 Histone modifications in the kiwifruit *AcFLCL* locus correlate with sense and antisense expression. (a) *AcFLCL* locus. Exons are presented as boxes, the translation start and stop positions are indicated above, primer positions and numbers are presented as orange triangles below. (b) The histone mark H3K4me3 abundance in winter (9 July) and before budbreak (16 August). (c) Accumulation of *AcFLCL* sense and antisense transcripts in winter and before budbreak. (d) The histone mark H3K4me3 in buds exposed to cold before treatment (Day 0) and after 36 h (Day 1.5) and 30 d (Day 30) of cold treatment. (e) Expression of sense and antisense transcripts at the same time points. (f) The histone mark H3K27me3 in the same set of bud samples. Quantification of histone modifications was performed by chromatin immunoprecipitation coupled with quantitative polymerase chain reaction (ChIP-qPCR) and presented as percentage input, with error bars representing ± SE of three biological replicates (b, d, f). Accumulation of sense and antisense transcripts were presented as a mean ± SE of three biological replicates relative to *ACTIN* (c, e).

repeated exons 6 and 7 (g4) (Fig. 7a) was performed. A total of 20 transgenic lines were produced, including five with confirmed bi-allelic edits, and two with confirmed edits in one allele, while one or both of the PCR products of the other allele failed to be amplified, suggesting the presence of larger rearrangements (Fig. 7b). Six lines with bi-allelic edits and six nonedited lines were grown in the glasshouse. Following defoliation in early winter, all lines remained dormant during the coldest period of the year. The first budbreak was recorded in control lines in early August, and all control lines broke buds within the period of 15 d. Edited lines started breaking buds 20 d after nonedited lines. A delay of 22 d was recorded between the time when all control and all edited lines broke buds (Fig. 7c). A delay in budbreak was again observed in the following year in the glasshouse-grown lines exposed to ambient winter temperatures (Fig. 7d) and using a forced budbreak assay with samples collected before substantial winter chilling (29 June) from three nonedited control lines and five edited lines. The controls started breaking buds within 16–19 d and most were growing shoots within 4 wk. In contrast, the first budbreak in edited lines ranged from 21 to 30 d and a large proportion of buds remained dormant after 4 wk (Fig. 7e,f).

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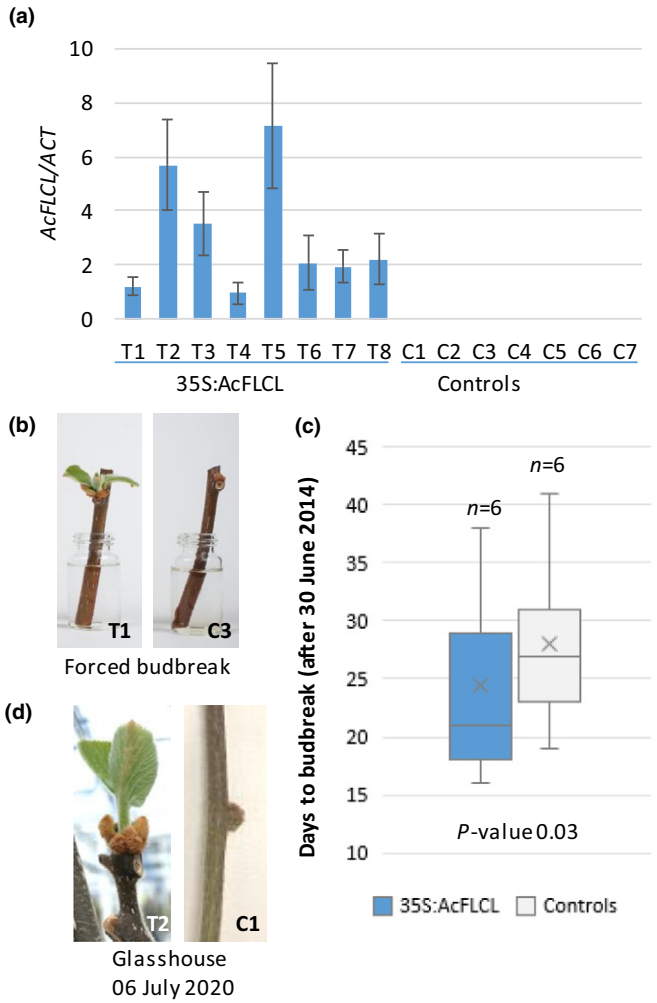


Fig. 6 *AcFLCL* advances the timing of kiwifruit budbreak. (a) Relative expression of *AcFLCL* in leaves of transgenic *35S:AcFLCL* lines (T) and control *35S:GUS* (C) lines of kiwifruit *Actinidia chinensis*. The expression was normalized to kiwifruit *ACTIN* and presented as a mean ± SE of three technical replicates. (b) Forced budbreak assay on cuttings collected on 30 June 2014. (c) The box plots representing the number of days to visible budbreak of six independent *35S:AcFLCL* and six control *35S:GUS* lines. The whiskers represent the minimum and maximum, the bottom and top of the box are the 25th and 75th percentiles, the line inside the box is the 50th percentile (median) and the cross represents the mean. A minimum of five cuttings per line was used in a forced budbreak assay. (d) Example of an early shoot outgrowth during winter (6 July 2020) in the glasshouse-grown transgenic *35S:AcFLCL* lines. The first budbreak in controls was recorded in August.

Discussion

Transcriptomics of endodormancy and chill responses

Dormancy ensures survival of winter conditions and is essential for kiwifruit flowering and yield (Snelgar *et al.*, 2007). The developmental and physiological changes that occur during dormancy establishment and release correspond with major transcriptome changes and reprogramming of many pathways. Study of buds exposed to field conditions identified genes known to perform important regulatory roles in temperate species. For example, a *NINE-CIS-EPOXYCAROTENOID DIOXYGENASE 3* (*NCED3*) homolog Acc21754 peak expression in May coincided with leaf drop, consistent with its rate-limiting role in synthesis of the dormancy-inducing agent, abscisic acid (ABA). It also coincided with elevated expression of a kiwifruit dormancy marker *DORMANCY-ASSOCIATED PROTEIN 1* (*DRM1*) (Wood *et al.*, 2013). The growth and flowering-related genes were upregulated in late winter and at budbreak. For example, within the set of genes showing upregulation between May and July, *AcFT* showed the largest fold change, whilst upregulation of *FUL* homologs indicated resumption of growth and flowering.

In contrast to the field experiment, where cold periods were interrupted by mid-winter warm spells and the dormancy status was a function of accumulated chilling units, the second approach studied responses of buds on excised canes to continual chilling. The lack of warm periods prevented developmental and physiological changes accompanying initiation of growth, allowing us to monitor genes and pathways affected by cold. Many genes were shown to be differentially expressed in both experiments. Elevated expression of genes regulating responses to multiple stresses, reactive oxygen species (ROS), sequestering of iron, and accumulation of osmoprotectants all probably contributed to cold acclimation and stress tolerance. Dormancy release was accompanied by reduced expression of many of these genes, although many remained upregulated with continual chilling. Surprisingly, accumulation of flowering-related genes was detectable after a short exposure to persistent cold, well below the chilling requirement of kiwifruit. This is consistent with the lower chilling requirement for flowering reported for some tree species, e.g. pear *Pyrus pyrifolia* (Hussain *et al.*, 2015), but in contrast to a study in kiwifruit that showed a higher chilling requirement for floral emergence than for vegetative budbreak (Tingting *et al.*, 2017).

FLC-like genes and bud phenology

MADS-box genes with similarity to Arabidopsis *FLC* have been identified in many plants, although it remains to be proven that any are true *FLC* or *MAF* orthologs. It was proposed that they arose with *AP1*/*FUL* genes through tandem duplication of a common ancestral gene, but have been lost or undergone structural changes in different angiosperm lineages (Ruelens *et al.*, 2013; Yu *et al.*, 2016). Genomic sequencing and large-scale phylogenetic studies of tree species identified MADS-box genes in the same subclade as Arabidopsis *FLC* and *MAF*, justifying the *FLC*-like annotation, but none of the sequences was similar enough to be regarded as an individual *FLC* ortholog. Similarly, although kiwifruit FLC-like predicted proteins displayed more similarity to FLC and MAF proteins than to any of the other MADS-box classes in Arabidopsis, they phylogenetically form separate subclades. Clearly, multiple independent expansions occurred within FLC/MAF lineages, giving rise to distinct subclades specific to lower rank taxa, in a manner comparable to *DAM* and *SVP* lineages (Wells *et al.*, 2015; Liu *et al.*, 2018). The presence of two highly similar and one more divergent kiwifruit *FLC*-like gene reflects the gene duplication events reported for *Actinidia* (S. Huang *et al.*, 2013; Pilkington *et al.*, 2018).

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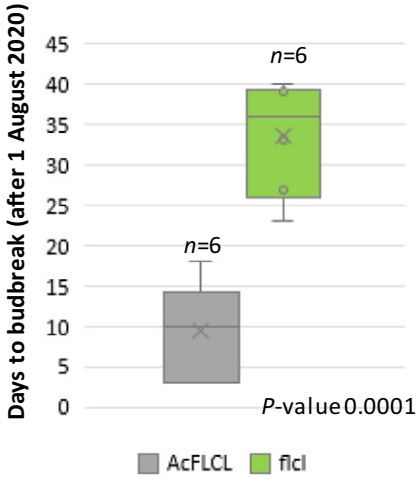
(a)



(b)

Line#	Allele	g1	g2	Target g3	g4a	g4b
18	1	WT	WT	**−8 bp**	WT	**+1 bp**
	2	WT	WT	**−1 bp**	WT	**+1 bp**
21	1	**+1 bp**	WT	**−8 bp**	WT	WT
	2	WT	WT	**−9 bp**	WT	WT
23	1	WT	WT	**−1 bp**	**+1 bp**	WT
	2	ND	ND	ND	WT	WT
24	1	WT	WT	**−5 bp**	WT	WT
	2	WT	WT	WT	WT	WT
25	1	**−2 bp**	WT	WT	**Δ-g4b**	**Δ-g4a**
	2	**+1 bp**	WT	**inv-g4b**	**+1 bp**	**−1 bp + inv-g3**
28	1	WT	WT	WT	**+1 bp**	WT
	2	WT	WT	WT	WT	WT
32	1	**−4 bp + inv-g3**	**inv-g4b**	**inv-g1**	**−1 bp**	**+1 bp + inv-g2**
	2	ND	ND	ND	**Δ-g4b**	**+1 bp + Δ-g4a**
34	1	**−3 bp**	WT	**−1 bp**	ND	**+1 bp**
	2	**−7 bp**	WT	**−10 bp**	**+1 bp**	WT
35	1	WT	WT	**−46 bp**	WT	**+1 bp**
	2	ND	ND	ND	ND	ND

(c)



(d)



(e)



(f)

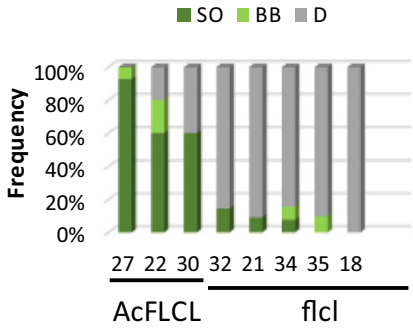


Fig. 7 Mutagenesis of *AcFLCL* delays budbreak in kiwifruit. (a) Design of gene editing and genotyping; fragment of the *AcFLCL* locus with exons in green, arrows above indicating the positions of targets for gene editing (g1–g4), forward (for) and reverse (rev) primers for genotyping, and PCR fragments for sequencing indicated as purple boxes. (b) Mutations in transgenic lines. ND, not determined; WT, wild-type allele; Δ, deletion; inv, inversion. Lines 18, 21, 25, 32 and 34 have bi-allelic edits; lines 23 and 35 have confirmed edits in one allele and probable large rearrangements in the other allele which failed to be amplified. (c) The timing of first budbreak measured as days after 1 August 2020 in nonedited controls (AcFLCL) and edited lines (flcl), represented as box plots of six independent AcFLCL and six flcl lines. The whiskers represent the minimum and maximum, the bottom and top of the box are the 25th and 75th percentiles, the line inside the box is the 50th percentile (median) and the cross represents the mean. (d) Delayed budbreak in glasshouse-grown plants (photographs taken on 10 August 2021). (e) Dormant (D), budbreak (BB) and shoot outgrowth (SO) stages in the forced budbreak assay. (f) Delayed budbreak in edited lines recorded after 4 wk in the forced budbreak assay, using a minimum of 10 cuttings per line. The sampling was performed on 29 June 2021 and the graph represents the scoring on 26 July 2021.

AcFLCL was elevated during winter, coinciding with peak chilling accumulation. The upregulation of *FLC* expression in buds concomitant with exposure to low temperature has been found in other perennial species, including apple (Porto *et al.*, 2015; Nishiyama *et al.*, 2021) and citrus (Zhang *et al.*, 2009). In poplar, opposing expression patterns were reported for *PtFLC2* and *PtFLC4* (Chen *et al.*, 2018). *PtFLC2* expression decreased in response to chilling during winter dormancy and increased after dormancy release, resembling expression of *VvFLC1* in grape and the *FLC* homolog *PEP1* in the perennial Arabidopsis relative *Arabis alpina* (Wang, 2009; Soppe *et al.*, 2021). Conversely, *PtFLC4* expression increased during dormancy and declined with resumption of growth in spring. This was similar to grape *VvFLC2* and an apple *MdFLC* (Wang, 2009; Díaz-Riquelme *et al.*, 2012; Kumar *et al.*, 2016; Chen *et al.*, 2018), as has been now demonstrated for *AcFLCL*, suggesting a potentially conserved role in bud phenology. Upregulation of *AcFLCL* during early winter and further increase with accumulation of winter chilling is suggestive of multiple roles this gene may perform during establishment and release of dormancy. Alternatively, it is possible that a threshold of expression may be required for function, or that AcFLCL function depends on interactions with stage-specific protein partners.

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FLC-like genes can regulate growth

While *FLC* is the key regulator of vernalization-mediated flowering, it remains unclear if *FLC*-like genes have the same role in species outside the Brassicaceae. A vernalization-responsive *FLC* homolog in the biennial sugar beet showed no role in flowering (Reeves *et al.*, 2007; Vogt *et al.*, 2014), however an *FLC* homolog from root chicory could delay flowering in transgenic Arabidopsis (Périlleux *et al.*, 2013). Very little is known about the role of *FLC*-like genes in perennials, other than *PEP1* in *Arabis alpina*, shown to limit the duration of flowering, maintain vegetative branches, and regulate seasonal flowering (Wang, 2009; Soppe *et al.*, 2021).

The functional analysis of woody perennial *FLC*-like genes has been mostly limited to model plants. An apple *FLC* (MD09G1009100) inhibited growth and delayed bolting in Arabidopsis (Nishiyama *et al.*, 2021). *MdFLC3* (MD10G1041100) also acted as a floral repressor in Arabidopsis (Kagaya *et al.*, 2020). *MdFLC3* showed reduced expression during winter dormancy and elevated expression after dormancy release, similar to *PtFLC2*, *VvFLC1* and a highly homologous pear *FLC* (Niu *et al.*, 2015). When constitutively expressed in transgenic blueberries, *MdFLC3* induced flowering without chilling (Zong *et al.*, 2019). The discrepancy in function may indicate a difficulty with annotation of MADS-box genes. *MdFLC3* was 88% identical to pear *FLC*, but also showed 82% identity to the blueberry *CAULIFLOWER* homolog *VcCAL1*, and was clearly performing as a functionally conserved flowering activator in blueberry (Zong *et al.*, 2019).

AcFLCL shows homology to *AP1*/*CAL*/*FUL* but does not have the specific C-terminal signature sequences, in contrast to true kiwifruit orthologs of these genes (Varkonyi-Gasic *et al.*, 2011). Overexpression of *AcFLCL* in *Actinidia chinensis* resulted in advanced shoot outgrowth, suggesting a growth promotion role. This was proven beyond doubt by generating mutations in the *AcFLCL*. Bi-allelic mutations were clearly correlated with a delay in visible budbreak. We conclude that *AcFLCL* accumulation during winter contributes to dormancy release, acting to enable shoot outgrowth. This effect is comparable to Arabidopsis *FLC* underlying the *REDUCED STEM BRANCHING6* (*RSB6*) locus (X. Huang *et al.*, 2013). Another pleotropic effect of Arabidopsis *FLC* is regulation of temperature-mediated seed germination (Chiang *et al.*, 2009; Chen *et al.*, 2014; Auge *et al.*, 2017; Blair *et al.*, 2017). This is of interest because aspects of the physiology and regulatory mechanisms of seed and bud dormancy may be shared (Leida *et al.*, 2012a; Beauvieux *et al.*, 2018). *FLC*-mediated promotion of germination is dependent on *FT*, but in a pathway with an opposite *FT*–*FLC* interaction. Unlike in regulation of flowering where FLC represses *FT* to inhibit flowering, FT represses *FLC* via activation of antisense *FLC* transcription and by interaction with a 5′ *FLC* promoter region (Chen & Penfield, 2018; Luo *et al.*, 2019). The *FLC*-mediated germination is similar to our finding that *AcFLCL* promotes budbreak, but at this stage it is unclear if an interaction with an *FT* gene is involved. AcFLCL may exert its function through activation of different genes and pathways, leading to a steady increase in cell division and expansion genes that precedes visible budbreak and shoot outgrowth. Enrichment for CArG motifs in promoters of co-expressed genes indicates possible multiple targets, including other MADS-box genes shown to regulate phenology in other plant species. For example, activation of *AP1*/*FUL* genes may provide a pathway similar to *LAP1*, shown to mediate seasonal growth in poplar (Azeez *et al.*, 2014).

In addition to the effect on vegetative phenology, manipulation of *AcFLCL* may cause pleiotropic effects in kiwifruit transgenic lines. For example, the loss of *PEP1* in *Arabis alpina* reduced the seed dormancy through an altered ABA response (Hughes *et al.*, 2019). We could not yet determine if a mutation in *AcFLCL* impacts on kiwifruit seed germination, or other reproductive traits. A potential effect on flower development and flower numbers could not be evaluated because the plants have not yet reached maturity. However, we could demonstrate that *AcFLCL* has no major effect on transition to reproductive maturity in kiwifruit, in contrast to Arabidopsis *FLC* (Deng *et al.*, 2011).

Epigenetics – a window into evolution and functional diversification of *FLC* homologs?

In addition to the long sense transcript, an antisense transcript with an opposite expression and small sense transcripts have been detected in the *AcFLCL* locus, resembling the Arabidopsis *COOLAIR* and *COLDWRAP* regulatory ncRNAs. This has not been observed with other MADS-box genes in kiwifruit buds. Such ncRNAs have not yet been reported for *FLC* homologs outside the Brassicaceae and it remains unclear if they perform a regulatory role in kiwifruit, but this study suggests that they may be a conserved feature. This scenario is consistent with genes arising from the same common ancestor, evolving to acquire different functions in plants with different life histories, but retaining aspects of regulation to a common stress, cold exposure. The chilling response has been proposed to have derived from the drought stress response, and they share many commonalities (Zhu, 2016; Hussain *et al.*, 2018; X. Ma *et al.*, 2020). Vernalization-mediated flowering in Arabidopsis can also be interpreted as dehydration avoidance and drought escape (McKay *et al.*, 2003), explaining similar genetic regulation involving a conserved *FLC*-like regulatory module. Furthermore, histone modification was identified in the *AcFLCL* locus. This type of regulation may be more common for MADS-box genes and has been reported for another regulator of kiwifruit phenology, *SVP2* (Wu *et al.*, 2019), as well as *DAM*/*SVP* genes in several horticultural trees in the Rosaceae family (Leida *et al.*, 2012b; de la Fuente *et al.*, 2015; Niu *et al.*, 2015; Saito *et al.*, 2015; Rothkegel *et al.*, 2017; Zhu *et al.*, 2020). In the *AcFLCL* locus, dynamic changes in H3K4me3 activation mark were detected, although *AcFLCL* did not appear to be subject to repression by H3K27me3 histone modifications, unlike Arabidopsis *FLC* and citrus floral repressor *CcMADS19* (Hou *et al.*, 2014; Agustí *et al.*, 2020), but similar to kiwifruit *SVP* and poplar *SVL* (Singh *et al.*, 2018; Wu *et al.*, 2019). Therefore, it is possible that some but not all aspects of epigenetic regulation play a universally important role in regulation of MADS-box genes associated with

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dormancy and chilling, opening new avenues for bioengineering and for future breeding strategies, with implications for food security in the changing climate.

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Author contributions

CV and EV-G conceived the study and designed the experiments. CV, LAB, RW and EV-G conducted the experiments. CV and TW generated and maintained the transgenic lines. CV, LAB and EV-G analysed the data. EV-G, CV and RW drafted the manuscript and ACA revised the manuscript. All authors approved the manuscript.

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Data availability

The data that supports the findings of this study are available in the Supporting Information of this article.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Daily temperatures during bud sampling.

Fig. S2 *Actinidia chinensis* bud transcriptome changes over the season and after cold treatment.

Fig. S3 Gene ontology analysis.

Fig. S4 Gene ontology analysis.

Fig. S5 Phylogenetic analysis of annotated MADS-box predicted proteins from *Actinidia chinensis* with *Arabidopsis thaliana* MADS-box proteins.

Fig. S6 Alignment of kiwifruit FLC-like and *Arabidopsis* FLC/MAF predicted proteins.

Fig. S7 Expression of kiwifruit MADS-box gene homologs in axillary buds.

Table S1 Oligonucleotide primers used in this study.

Table S2 Gene editing design and cloning of the construct.

Table S3 Differentially expressed genes in axillary buds identified by DESeq.

Table S4 Differentially expressed genes in axillary buds of field-grown plants.

Table S5 Differentially expressed genes in axillary buds on excised canes exposed to chilling.

Table S6 Differentially expressed genes associated with bud dormancy onset and release in the field and in response to cold.

Table S7 Differentially expressed genes relevant for growth resumption and budbreak.

Table S8 Overrepresentation or underrepresentation of CArG box motifs in the promoters of differentially expressed genes.

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